



The logo for VIT (Vellore Institute of Technology) features a solid red horizontal line positioned above the letters "VIT", which are rendered in a bold, black, sans-serif typeface.The logo for FACE (Fellowship for Advanced Computing and Engineering) consists of a red icon on the left, composed of several small squares arranged in a stepped, staircase-like pattern, followed by the word "FACE" in a bold, black, sans-serif font.

# Euclidean Algorithm

- ❖ Euclidean Algorithm or Euclid's Algorithm.
- ❖ For computing the Greatest Common Divisor (GCD).

## Understanding GCD – Example 2

|                               | 25       | 150                                        |
|-------------------------------|----------|--------------------------------------------|
| Divisors                      | 1, 5, 25 | 1, 2, 3, 5, 6, 10, 15, 25, 30, 50, 75, 150 |
| Common Divisors               | 1, 5, 25 |                                            |
| Greatest Common Divisor (GCD) | 25       |                                            |

$$\therefore \text{GCD}(25, 150) = 25$$

## Understanding GCD – Example 3

|                               | 13    | 31    |
|-------------------------------|-------|-------|
| Divisors                      | 1, 13 | 1, 31 |
| Common Divisors               | 1     |       |
| Greatest Common Divisor (GCD) | 1     |       |

$$\therefore \text{GCD}(13, 31) = 1$$

# Euclid's Algorithm for finding GCD

$$\text{GCD}(12, 33) = 3.$$

| Q | A  | B  | R |
|---|----|----|---|
| 2 | 33 | 12 | 9 |
| 1 | 12 | 9  | 3 |
| 3 | 9  | 3  | 0 |
| X | 3  | 0  | X |

# Euclid's Algorithm for finding GCD

Find the GCD(750, 900).

| Q | A   | B   | R   |
|---|-----|-----|-----|
| 1 | 900 | 750 | 150 |
| 5 | 750 | 150 | 0   |
| X | 150 | 0   | X   |

The diagram illustrates the steps of Euclid's Algorithm for finding the GCD of 750 and 900. The table shows the sequence of divisions:

- Step 1:  $900 = 1 \times 750 + 150$
- Step 2:  $750 = 5 \times 150 + 0$
- The process ends when the remainder is 0, and the last non-zero remainder is 150.

The final result, 150, is highlighted by a red dashed box, indicating it is the GCD.

# Euclidean Algorithm

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## Euclid's Algorithm for finding GCD

Prerequisite:  $a > b$

Euclid\_GCD ( $a, b$ ):

    if  $b = 0$  then

        return  $a$ ;

    else

        return Euclid\_GCD ( $b, a \bmod b$ );

# Euclid's Algorithm – Example 1

Example 1: Find the GCD (50, 12).

Solution:

Here  $a=50$ ,  $b=12$

$$\text{GCD}(a, b) = \text{GCD}(b, a \bmod b)$$

$$\text{GCD}(50, 12) = \text{GCD}(12, 50 \bmod 12) = \text{GCD}(12, 2)$$

$$\text{GCD}(12, 2) = \text{GCD}(2, 12 \bmod 2) = \text{GCD}(2, 0) = 2$$

$$\text{GCD}(50, 12) = 2$$



# Euclid's Algorithm – Example 2

Example 2: Find the GCD (83, 19).

Solution:

Here  $a=83$ ,  $b=19$

$$\text{GCD}(a, b) = \text{GCD}(b, a \bmod b)$$

$$\text{GCD}(83, 19) = \text{GCD}(19, 83 \bmod 19) = \text{GCD}(19, 7)$$

$$\text{GCD}(19, 7) = \text{GCD}(7, 19 \bmod 7) = \text{GCD}(7, 5)$$

$$\text{GCD}(7, 5) = \text{GCD}(5, 7 \bmod 5) = \text{GCD}(5, 2)$$

$$\text{GCD}(5, 2) = \text{GCD}(2, 5 \bmod 2) = \text{GCD}(2, 1)$$

$$\text{GCD}(2, 1) = \text{GCD}(1, 2 \bmod 1) = \text{GCD}(1, 0) = 1$$

$$\text{GCD}(83, 19) = 1$$

```
import java.util.*;
```

```
3
```

```
4 public class GFG {
```

```
5
```

```
6     public static int gcd(int a, int b)
```

```
7     {
```

```
8         if (a == 0)
```

```
9             return b;
```

```
10
```

```
11         return gcd(b % a, a);
```

```
12     }
```

```
13
```

```
14     // Driver code
```

```
15     public static void main(String[] args)
```

```
16     {
```

```
17         int a = 50, b = 12, g;
```

```
18
```

```
19         // Function call
```

```
20         g = gcd(a, b);
```

```
21         System.out.println("GCD(" + a + " , " + b
```

```
22             + ") = " + g);
```

```
23
```

```
24     }
```

```
25 }
```